CiliaTracks

End-to-End Application Tutorial

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This tutorial presents the complete application-based CiliaTracks workflow. Detailed Python function documentation is provided in the CiliaTracks Instruction Manual.

1 Overview and workflow

CiliaTracks is an application and Python package for visualizing, quantifying, and classifying cilia-driven microsphere transport from high-speed video microscopy (HSVM). This worked example follows a control airway epithelial culture through biological preparation, video acquisition, automated TrackMate analysis, and generation of a single Quick Summary report.

The application accepts the matching track_statistics and spot_statistics CSV files generated from one recording. Predictions are produced at the recording level and should be reviewed across replicate recordings and biological context.

End-to-end workflow

1. Prepare a mature, adequately ciliated control air-liquid interface (ALI) culture.
2. Apply fluorescent microspheres and capture standardized high-speed videos.
3. Run the provided automated Fiji/ImageJ TrackMate script.
4. Upload the matching Tracks and Spots CSV files to Quick Summary.
5. Review both model predictions, plots, and extracted features.

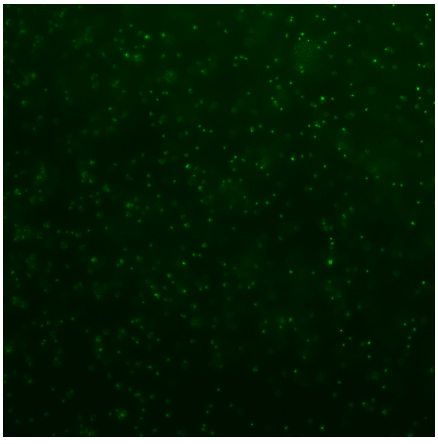
2 Biological preparation for control cultures

2.1 Materials

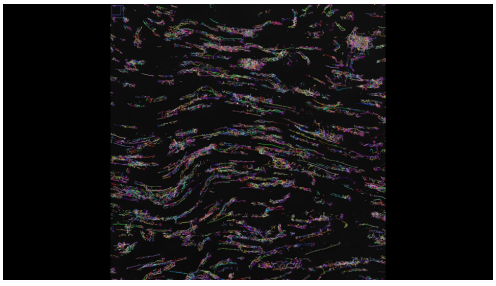
- 1% solids, 1.9 µm green fluorescent polymer microspheres (Thermo Fisher Scientific, G0200).
- Warm 2% Nu-Serum medium (Corning, 355100).
- Mature control airway epithelial cells differentiated on 6.5 or 12 mm Transwell inserts.
- Widefield inverted fluorescence microscope capable of recording at 100 frames per second.

2.2 Microsphere application

1. Dilute the microspheres 1:1000 in warm 2% Nu-Serum and incubate at 37 °C for at least 10 minutes.
2. Gently wash the apical surface with warm 2% Nu-Serum for approximately 1 minute, then aspirate the medium.
3. Mix and slowly apply the microsphere suspension to the center of the insert: 20 µL for a 6.5 mm insert or 60 µL for a 12 mm insert.
4. Incubate the culture for 10 minutes at 37 °C before imaging.



Fluorescent microspheres before tracking.

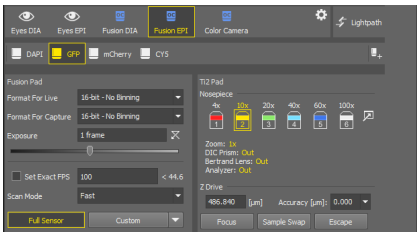


Representative organized control particle movement.

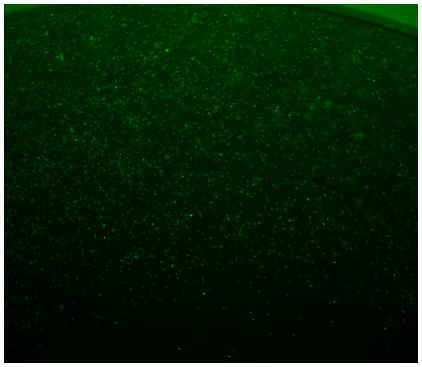
3 Capture high-speed videos

Use the standardized acquisition conditions below. In the study, reduced recording temperature and inadequate ciliation decreased classification performance; recordings should therefore be collected at 37 °C from well-ciliated cultures.

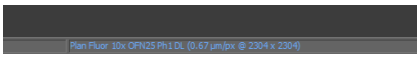
1. Stabilize the microscope and environmental chamber at 37 °C.
2. Use the GFP fluorescence channel and a 10× objective.
3. Select a well-focused field with visible microsphere movement. Suitable regions are often away from the insert center.
4. Capture a 4.45-second recording at 100 frames per second (445 frames).
5. Record the acquisition calibration in µm/pixel and save the video as AVI with MJPEG compression when available.
6. Capture at least three recordings from different regions of each culture insert.



Representative GFP and 10× acquisition settings.



Select a field with visible bead movement and adequate ciliation.



Record the µm/pixel calibration for the selected resolution.

Important: Aim for approximately 80% or greater surface ciliation for control cultures, maintain 37 °C during recording, and keep frame rate, magnification, resolution, exposure, and bead preparation consistent across the dataset.

After imaging, rinse the insert three times with warm 2% Nu-Serum if it will be retained for subsequent analysis.

4 Generate TrackMate CSV files in Fiji/ImageJ

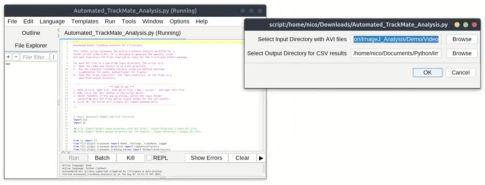
Use the provided automated CiliaTracks Jython TrackMate script as the default tracking workflow. The script batch-processes AVI files, converts each recording to 8-bit grayscale, runs the Laplacian of Gaussian detector and Kalman tracker, and exports the two CSV files required by the application.

4.1 Run the automated script

1. Download Automated_Trackmate_Analysis.py from the repository Fiji_Scripts folder on GitHub.
2. In Fiji/ImageJ, select File > New > Script and open the downloaded script.
3. Click Run, select the folder containing the AVI files, and choose a separate output folder.
4. Click OK. The script processes all AVI files and exports one matching CSV pair per recording.



Download the automated script from the repository.



Run the script and select the AVI input and CSV output folders.

4.2 Default settings and simple adjustment

TrackMate parameter	Script default
RADIUS	3.5 pixels
THRESHOLD	1.0
LINKING_MAX_DISTANCE	15 pixels
KALMAN_SEARCH_RADIUS	20 pixels
MAX_FRAME_GAP	1 frame

If the experimental setup differs from the validated widefield workflow, first run the default script on one representative recording and inspect the results. Make only limited changes:

- If visible beads are missed or background is over-detected, adjust RADIUS for particle size/resolution and THRESHOLD for signal intensity.
- If tracks fragment, adjust KALMAN_SEARCH_RADIUS or MAX_FRAME_GAP; if tracks jump between unrelated beads, reduce the linking/search distance.
- For confocal recordings, THRESHOLD = 3.5 and MAX_FRAME_GAP = 6 can be used as starting values.
- Change one parameter at a time, document the final settings, and apply the same settings consistently within a dataset.

The script exports track_statistics_<video>.csv and spot_statistics_<video>.csv. Upload only the matching files generated from the same AVI recording.

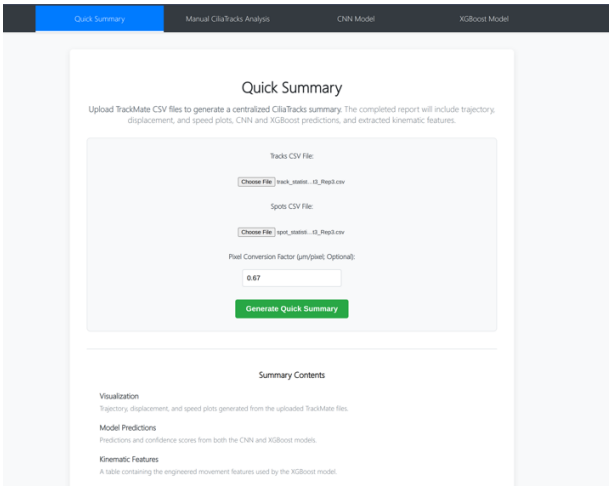
Full TrackMate documentation is available here. It is entirely justifiable to follow TrackMate's documentation and conduct a manual GUI-based analysis to generate the output files.

5 Generate the Quick Summary report

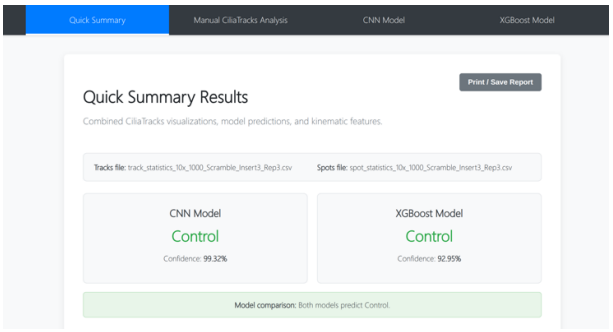
After TrackMate generates the matching CSV pair, the Quick Summary tab completes all downstream CiliaTracks steps in a single action.

5.1 Upload the TrackMate output files

1. Launch the CiliaTracks application and open the Quick Summary tab.
2. Upload the matching track_statistics and spot_statistics CSV files.
3. Enter the acquisition conversion factor in µm/pixel.
4. Click Generate Quick Summary.



Upload the matching CSV pair and µm/pixel conversion factor.



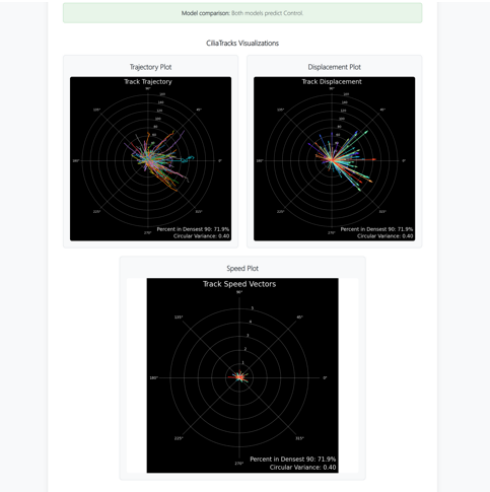
Worked control example: both models classify the recording as Control.

5.2 Review the automated output

- CNN and XGBoost predictions with confidence scores and a model-agreement indicator.
- Trajectory, displacement, and speed plots.
- Twelve extracted kinematic features with units.
- A printable report that can be saved through the browser print dialog.

6 Interpret and quality-check the output

In this worked example, both models classify the recording as Control. The trajectory and displacement plots show organized directional transport with substantial displacement, while the speed plot summarizes straight-line speed vectors. The extracted feature table reports the corresponding quantitative values.



Trajectory, displacement, and speed plots in the Quick Summary report.

XGBoost Kinematic Features

Feature	Value	Unit
Track displacement	36.1247	µm
Track mean speed	0.4107	µm/frame
Track max speed	3.9908	µm/frame
Track min speed	0.0547	µm/frame
Total distance traveled	60.0823	µm
Max distance traveled	36.9151	µm
Confinement ratio	0.6110	Unitless
Mean straight line speed	0.2321	µm/frame
Linearity of forward progression	0.6147	Unitless
Mean directional change rate	0.8908	radians/frame
Circular variance	0.4040	Unitless
Percent in densest 90°	71.8750	%

Extracted XGBoost features.

6.1 Final review checklist

- Confirm that the Tracks and Spots files were generated from the same recording.
- Confirm that the µm/pixel conversion matches the acquisition resolution.
- Repeat the analysis for at least three recordings from different regions of the culture insert.
- If the models disagree or replicate predictions are inconsistent, review culture ciliation, recording temperature, image quality, and TrackMate settings before interpretation.

6.2 Interpretation and troubleshooting

Issue	Recommended check
Few or no detected spots	Review focus and fluorescence signal; adjust THRESHOLD or RADIUS on a representative video.
Fragmented tracks	Cautiously increase the Kalman search radius or maximum frame gap.
Implausible track jumps	Reduce the linking/search distance and confirm that tracks remain associated with the same bead.
Inconsistent predictions	Confirm 37 °C acquisition, adequate ciliation, matching CSV files, calibration, and replicate consistency.

Interpretation: CiliaTracks predictions are generated for individual recordings and should not be treated as a stand-alone clinical diagnosis.